15.3 SCHMITT TRIGGER CIRCUITS

Objective: • Analyze and design various Schmitt trigger circuits.

In this section, we will analyze another class of circuits that utilize positive feedback. The basic circuit is commonly called a **Schmitt trigger,** which can be used in the class of waveform generators called multivibrators. The three general types of multivibrators are: bistable, monostable, and astable. In this section, we will examine the bistable multivibrator, which has a comparator with positive feedback and has two stable states. We will discuss the comparator first, and will then describe various applications of the Schmitt trigger.

15.3.1 Comparator

The comparator is essentially an op-amp operated in an open-loop configuration, as shown in Figure 15.24(a). As the name implies, a comparator compares two voltages

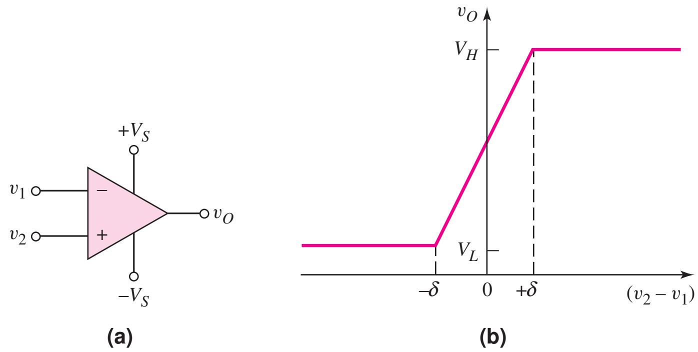

Figure 15.24 (a) Open-loop comparator and (b) voltage transfer characteristics, open-loop comparator

to determine which is larger. The comparator is usually biased at voltages $+V_S$ and $-V_S$, although other biases are possible.

The voltage transfer characteristics, neglecting any offset voltage effects, are shown in Figure 15.24(b). When v_2 is slightly greater than v_1, the output is driven to a high saturated state V_H; when v_2 is slightly less than v_1, the output is driven to a low saturated state V_L. The saturated output voltages V_H and V_L may be close to the supply voltages $+V_S$ and $-V_S$, respectively, which means that V_L may be negative. The transition region is the region in which the output voltage is in neither of its saturation states. This region occurs when the input differential voltage is in the range $-\delta < (v_2 - v_1) < +\delta$. If, for example, the open-loop gain is 10^5 and the difference between the two output states is $(V_H - V_L) = 10$ V, then

$$2\delta = 10/10^5 = 10^{-4}\text{ V} = 0.1\text{ mV}$$

The range of input differential voltage in the transition region is normally very small.

One major difference between a comparator and op-amp is that a comparator need not be frequency compensated. Frequency stability is not a consideration since the comparator is being driven into one of two states. Since a comparator does not contain a frequency compensation capacitor, it is not slew-rate-limited by the compensation capacitor as is the op-amp. Typical response times for the comparator output to change states are in the range of 30 to 200 ns. An expected response time for a 741 op-amp with a slew rate of 0.7 V/μs would be on the order of 30 μs, which is a factor of 1000 times greater.

Figure 15.25 shows two comparator configurations along with their voltage transfer characteristics. In both, the input transition region width is assumed to be negligibly small. The reference voltage may be either positive or negative, and the output saturation voltages are assumed to be symmetrical about zero. The crossover voltage is defined as the input voltage at which the output changes states.

Two other comparator configurations, in which the crossover voltage is a function of resistor ratios, are shown in Figure 15.26. Input bias current compensation is also included in this figure. From Figure 15.26(a), we use superposition to obtain

$$v_+ = \left(\frac{R_2}{R_1 + R_2}\right)V_{REF} + \left(\frac{R_1}{R_1 + R_2}\right)v_I \tag{15.64}$$

The ideal crossover voltage occurs when $v_+ = 0$, or

$$R_2 V_{REF} + R_1 v_I = 0 \tag{15.65(a)}$$

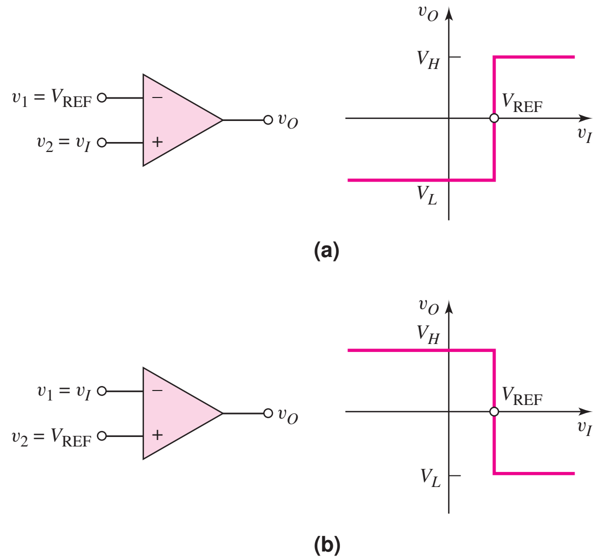

Figure 15.25 (a) Noninverting comparator circuit and (b) inverting comparator circuit

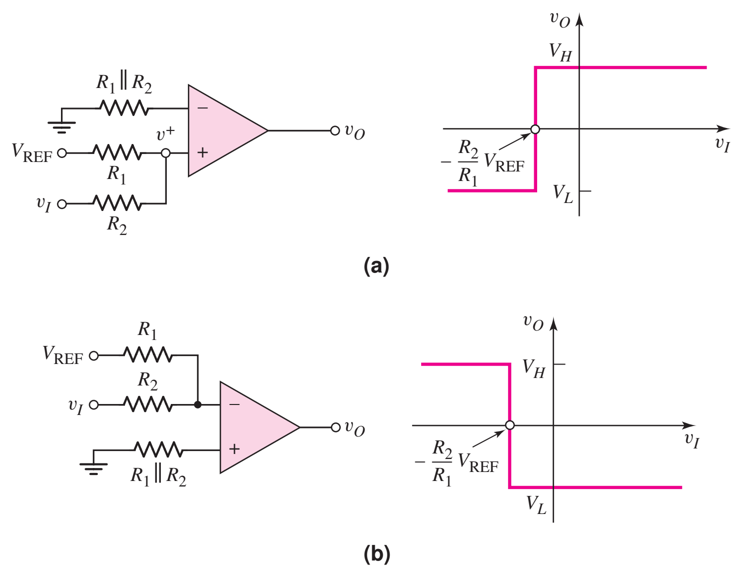

Figure 15.26 Other comparator circuits: (a) noninverting and (b) inverting

which can be written as

$$v_I = -\frac{R_2}{R_1} V_{REF} \tag{15.65(b)}$$

The output goes high when $v_+ > 0$. From Equation (15.64), we see that $v_o =$ High when v_I is greater than the crossover voltage. A similar analysis produces the characteristics shown in Figure 15.26(b).

Figure 15.27(a) shows one application of a comparator, to control street lights. The input signal is the output of a photodetector circuit. Voltage v_I is directly

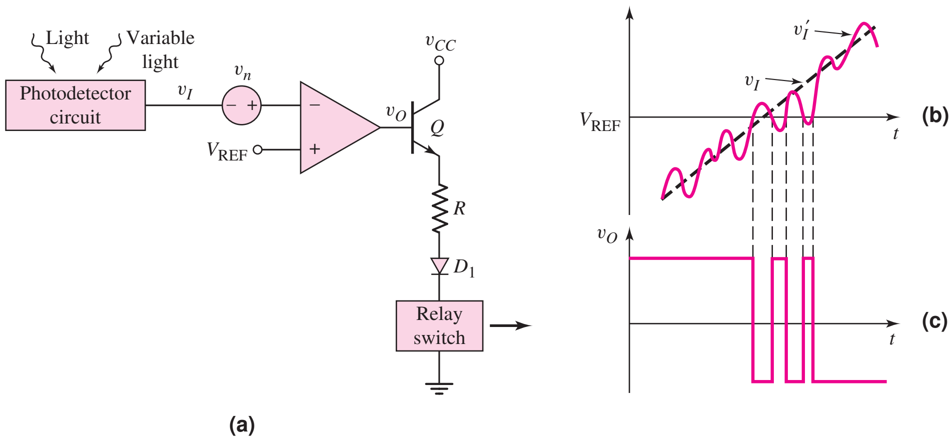

Figure 15.27 (a) Comparator circuit including input noise source, (b) input signal, and (c) output signal, showing chatter effect

proportional to the amount of light incident on the photodetector. During the night, $v_I < V_{REF}$, and v_O is on the order of $V_S = +15$ V; the transistor turns on. The current in the relay switch then turns the street lights on. During the day, the light incident on the photodetector produces an output signal such that $v_I > V_{REF}$. In this case, v_O is on the order of $-V_S = -15$ V, and the transistor turns off.

Diode D_1 is used as a protection device, preventing reverse-bias break-down in the B–E junction. With zero output current, the relay switch is open and the street lights are off. At dusk and dawn, $v_I = V_{REF}$.

The open-loop comparator circuit in Figure 15.27(a) may exhibit unacceptable behavior in response to noise in the system. Figure 15.27(a) shows the comparator circuit, with a variable light source, such as clouds causing the light intensity to fluctuate over a short period of time. A variable light intensity would be equivalent to a noise source v_n in series with the signal source v_I. If we assume that v_I is increasing linearly with time (corresponding to dawn), then the total input signal v_I' versus time is shown in Figure 15.27(b). When $v_I' > V_{REF}$, the output switches low; when $v_I' < V_{REF}$, the output switches high, producing a chatter effect in the output signal as shown in Figure 15.27(c). This effect would turn the street lights off and on over a relatively short time period. If the amplitude of the noise signal increases, the chatter effect becomes more severe. This chatter can be eliminated by using a Schmitt trigger.

15.3.2 Basic Inverting Schmitt Trigger

The Schmitt trigger or **bistable multivibrator** uses positive feedback with a loop-gain greater than unity to produce a bistable characteristic. Figure 15.28(a) shows one configuration of a Schmitt trigger. Positive feedback occurs because the feedback resistor is connected between the output terminal and noninverting input terminal. Voltage v_+, in terms of the output voltage, can be found by using a voltage divider equation to yield

$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right) v_O \tag{15.66}$$

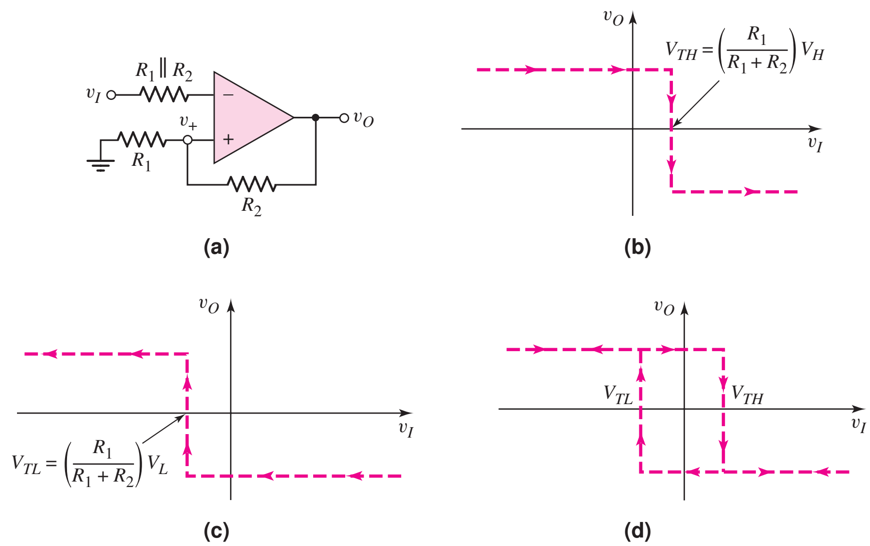

Figure 15.28 (a) Schmitt trigger circuit, (b) voltage transfer characteristic as input voltage increases, (c) voltage transfer characteristic as input voltage decreases, and (d) net voltage transfer characteristics, showing hysteresis effect

Voltage v_+ does not remain constant; rather, it is a function of the output voltage. Input signal v_I is applied to the inverting terminal.

Voltage Transfer Characteristics

To determine the voltage transfer characteristics, we assume that the output of the comparator is in one state, namely $v_O = V_H$, which is the high state. Then

$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right)V_H \tag{15.67}$$

As long as the input signal is less than v_+, the output remains in its high state. The crossover voltage occurs when $v_I = v_+$ and is defined as V_{TH}. We have

$$V_{TH} = \left(\frac{R_1}{R_1 + R_2}\right)V_H \tag{15.68}$$

When v_I is greater than V_{TH}, the voltage at the inverting terminal is greater than that at the noninverting terminal. The differential input voltage $(v_I - V_{TH})$ is amplified by the open-loop gain of the comparator, and the output switches to its low state, or $v_O = V_L$. Voltage v_+ then becomes

$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right)V_L \tag{15.69}$$

Since $V_L < V_H$, the input voltage v_I is still greater than v_+, and the output remains in its low state as v_I continues to increase. This voltage transfer characteristic is shown in Figure 15.28(b). Implicit in these transfer characteristics is the assumption that V_H is positive and V_L is negative.

Now consider the transfer characteristic as v_I decreases. As long as v_I is larger than $v_+ = [R_1/(R_1 + R_2)]V_L$, the output remains in its low saturation state. The crossover voltage now occurs when $v_I = v_+$ and is defined as V_{TL}. We have

$$V_{TL} = \left(\frac{R_1}{R_1 + R_2}\right)V_L \tag{15.70}$$

As v_I drops below this value, the voltage at the noninverting terminal is greater than that at the inverting terminal. The differential voltage at the comparator terminals is amplified by the open-loop gain, and the output switches to its high state, or $v_O = V_H$. As v_I continues to decrease, it remains less than v_+; therefore, v_O remains in its high state. This voltage transfer characteristic is shown in Figure 15.28(c).

Complete Voltage Transfer and Bistable Characteristics

The complete voltage transfer characteristics of the Schmitt trigger in Figure 15.28(a) combine the characteristics in Figures 15.28(b) and 15.28(c). These complete characteristics are shown in Figure 15.28(d). As shown, the crossover voltages depend on whether the input voltage is increasing or decreasing. The complete transfer characteristics therefore show a **hysteresis effect.** The width of the hysteresis is the difference between the two crossover voltages V_{TH} and V_{TL}.

The bistable characteristic of the circuit occurs around the point $v_I = 0$, at which the output may be in either its high or low state. The output remains in either state as long as v_I remains in the range $V_{TL} < v_I < V_{TH}$. The output switches states only if the input increases above V_{TH} or decreases below V_{TL}.

EXAMPLE 15.6

Objective: Determine the hysteresis width of a particular Schmitt trigger.

Consider the Schmitt trigger in Figure 15.28(a), with parameters $R_1 = 10\ \text{k}\Omega$ and $R_2 = 90\ \text{k}\Omega$. Let $V_H = 10$ V and $V_L = -10$ V.

Solution: From Equation (15.68), the upper crossover voltage is

$$V_{TH} = \left(\frac{R_1}{R_1 + R_2}\right)V_H = \left(\frac{10}{10 + 90}\right)(10) = 1\ \text{V}$$

and from Equation (15.70), the lower crossover voltage is

$$V_{TL} = \left(\frac{R_1}{R_1 + R_2}\right)V_L = \left(\frac{10}{10 + 90}\right)(-10) = -1\ \text{V}$$

The hysteresis width is therefore $(V_{TH} - V_{TL}) = 2$ V.

Comment: The hysteresis width can be designed to be larger or smaller for specific applications by adjusting the voltage divider ratio of R_1 and R_2.

EXERCISE PROBLEM

Ex 15.6: Consider the comparator circuit in Figure 15.28(a). Assume high and low saturated output voltages of +9 V and −9 V, respectively. Design the circuit such that the crossover voltages are ±0.5 V. The minimum resistance is to be 10 kΩ. (Ans. Set $R_1 = 10\ \text{k}\Omega$, then $R_2 = 170\ \text{k}\Omega$)

The complete voltage transfer characteristics in Figure 15.28(d) show the inverting characteristics of this particular Schmitt trigger. When the input signal becomes sufficiently positive, the output is in its low state; when the input signal is sufficiently negative, the output is in its high state. Since the input signal is applied to the inverting terminal of the comparator, this characteristic is as expected.

15.3.3 Additional Schmitt Trigger Configurations

A noninverting Schmitt trigger can be designed by applying the input signal to the network connected to the comparator noninverting terminal. Also, both crossover voltages of a Schmitt trigger circuit can be shifted in either a positive or negative direction by applying a reference voltage. We will study these general circuit configurations, the resulting voltage transfer characteristics, and an application of a Schmitt trigger circuit in this section.

Noninverting Schmitt Trigger Circuit

Consider the circuit in Figure 15.29(a). The inverting terminal is held essentially at ground potential, and the input signal is applied to resistor R_1, which is connected to the comparator noninverting terminal. Voltage v_+ at the noninverting terminal then becomes a function of both the input signal v_I and the output voltage v_O. Using superposition, we find that

$$v_+ = \left(\frac{R_2}{R_1 + R_2}\right)v_I + \left(\frac{R_1}{R_1 + R_2}\right)v_O \tag{15.71}$$

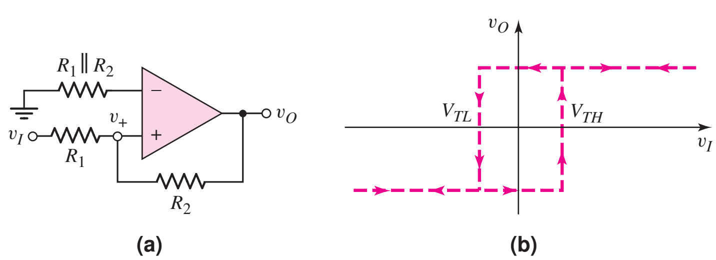

Figure 15.29 (a) Noninverting Schmitt trigger circuit and (b) voltage transfer characteristics

If v_I is negative, and the output is in its low state, then $v_O = V_L$ (assumed to be negative), v_+ is negative, and the output remains in its low saturation state. Crossover voltage $v_I = V_{TH}$ occurs when $v_+ = 0$ and $v_O = V_L$, or, from Equation (15.71),

$$0 = R_2 V_{TH} + R_1 V_L \tag{15.72(a)}$$

which can be written

$$V_{TH} = -\left(\frac{R_1}{R_2}\right)V_L \tag{15.72(b)}$$

Since V_L is negative, V_{TH} is positive.

If we let $v_I = V_{TH} + \delta$, where δ is a small positive voltage, the input voltage is just greater than the crossover voltage and Equation (15.71) becomes

$$v_+ = \left(\frac{R_2}{R_1 + R_2}\right)(V_{TH} + \delta) + \left(\frac{R_1}{R_1 + R_2}\right)V_L \tag{15.73}$$

Equation (15.73) then becomes

$$v_+ = \left(\frac{R_2}{R_1+R_2}\right)\left(\frac{-R_1}{R_2}\right)V_L + \left(\frac{R_2}{R_1+R_2}\right)\delta + \left(\frac{R_1}{R_1+R_2}\right)V_L \qquad \textbf{(15.74(a))}$$

or

$$v_+ = \left(\frac{R_2}{R_1+R_2}\right)\delta > 0 \qquad \textbf{(15.74(b))}$$

When $v_+ > 0$, the output switches to its high saturation state.

The lower crossover voltage $v_I = V_{TL}$ occurs when $v_+ = 0$ and $v_O = V_H$. From Equation (15.71), we have

$$0 = R_2 V_{TL} + R_1 V_H \qquad \textbf{(15.75(a))}$$

which can be written

$$V_{TL} = -\left(\frac{R_1}{R_2}\right)V_H \qquad \textbf{(15.75(b))}$$

Since $V_H > 0$, then $V_{TL} < 0$.

The complete voltage transfer characteristics are shown in Figure 15.29(b). We again note the hysteresis effect and the bistable characteristic around $v_I = 0$. With v_I sufficiently positive, the output is in its high state; with v_I sufficiently negative, the output is in its low state. The circuit thus exhibits the noninverting transfer characteristic.

Schmitt Trigger Circuits with Applied Reference Voltages

The switching voltage of a Schmitt trigger is defined as the average value of V_{TH} and V_{TL}. For the two circuits in Figure 15.28(a) and 15.29(a), the switching voltages are zero, assuming $V_{TL} = -V_{TH}$. In some applications, the switching voltage must be either positive or negative. Both crossover voltages can be shifted in either a positive or negative direction by applying a reference voltage.

Figure 15.30(a) shows an inverting Schmitt trigger with a reference voltage V_{REF}. The complete voltage transfer characteristics are shown in Figure 15.30(b). The switching voltage V_S, assuming V_H and V_L are symmetrical about zero, is given by

$$V_S = \left(\frac{R_2}{R_1+R_2}\right)V_{REF} \qquad \textbf{(15.76)}$$

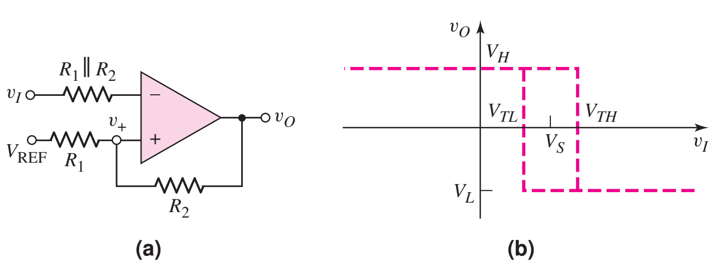

Figure 15.30 (a) Inverting Schmitt trigger circuit with applied reference voltage and (b) voltage transfer characteristics

Note that the switching voltage is not the same as the reference voltage. The upper and lower crossover voltages are

$$V_{TH} = V_S + \left(\frac{R_1}{R_1 + R_2}\right) V_H \qquad \textbf{(15.77(a))}$$

and

$$V_{TL} = V_S + \left(\frac{R_1}{R_1 + R_2}\right) V_L \qquad \textbf{(15.77(b))}$$

A noninverting Schmitt trigger with a reference voltage is shown in Figure 15.31(a), and the complete voltage transfer characteristics are shown in Figure 15.31(b). The switching voltage V_S, again assuming V_H and V_L are symmetrical about zero, is given by

$$V_S = \left(1 + \frac{R_1}{R_2}\right) V_{\text{REF}} \qquad \textbf{(15.78)}$$

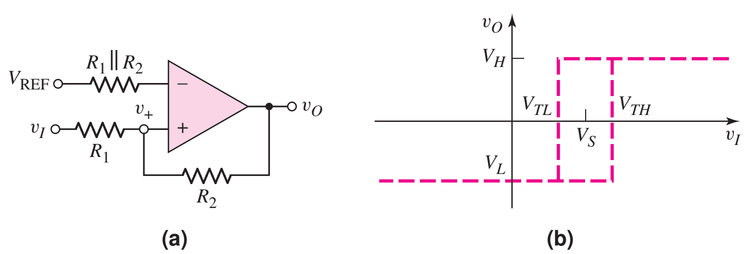

Figure 15.31 (a) Noninverting Schmitt trigger circuit with applied reference voltage and (b) voltage transfer characteristics

and the upper and lower crossover voltages are

$$V_{TH} = V_S - \left(\frac{R_1}{R_2}\right) V_L \qquad \textbf{(15.79(a))}$$

and

$$V_{TL} = V_S - \left(\frac{R_1}{R_2}\right) V_H \qquad \textbf{(15.79(b))}$$

If the output saturation voltages are symmetrical such that $V_L = -V_H$, then the crossover voltages are symmetrical about the switching voltage V_S.

Schmitt Trigger Application

Let us reconsider the street light control in Figure 15.27(a), which included a noise source. Figure 15.32(a) shows the same basic circuit, except that a Schmitt trigger is used instead of a simple comparator.

The input signal v_I is again assumed to increase linearly with time. The total input signal v_I' is v_I with the noise signal superimposed, as shown in Figure 15.32(b). At time t_1, the input signal becomes greater than the switching voltage V_S. The output, however, does not switch, since $v_I' < V_{TH}$. This means that the input signal is less than the upper crossover voltage. At time t_2, the input signal becomes larger than the crossover voltage, or $v_I' > V_{TH}$, and the output signal switches from its high to its low state. At time t_3, the input signal drops below V_S, but the output does not switch states since $v_I' > V_{TL}$. This means that the input signal remains greater than

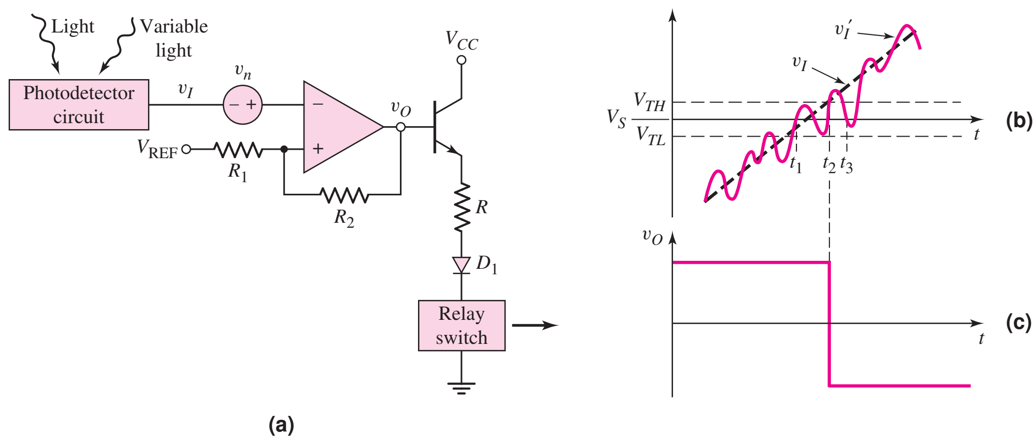

Figure 15.32 (a) Application of Schmitt trigger circuit including input noise source, (b) input signal, and (c) output signal, showing elimination of chatter effect

the lower crossover voltage. The Schmitt trigger circuit thus eliminates the chatter effect that occurs in the output voltage in Figure 15.27(c). Elimination of the chatter in the output voltage response results directly from the hysteresis effect in the Schmitt trigger characteristics.

DESIGN EXAMPLE 15.7

Objective: Design a Schmitt trigger circuit for the photodetector switch circuit.

Specifications: The Schmitt trigger circuit with the configuration shown in Figure 15.32(a) is to be designed such that the switching voltage is $V_S = 2$ V and the hysteresis width is 60 mV. Assume $V_H = 5$ V and $V_L = -5$ V.

Choices: An ideal comparator is available and standard-valued resistors are to be used in the final design.

Solution: The Schmitt trigger circuit is the inverting type, for which the voltage transfer characteristics are shown in Figure 15.30(b). From Equations (15.77(a)) and (15.77(b)), the hysteresis width is

$$V_{TH} - V_{TL} = \left(\frac{R_1}{R_1 + R_2}\right)(V_H - V_L)$$

so

$$0.060 = \left(\frac{R_1}{R_1 + R_2}\right)[5 - (-5)] = 10\left(\frac{R_1}{R_1 + R_2}\right)$$

which yields $R_2/R_1 = 165.7$. We can find the reference voltage from Equation (15.76), which can be rewritten to obtain

$$V_{REF} = \left(1 + \frac{R_1}{R_2}\right)V_S = \left(1 + \frac{1}{165.7}\right)(2) = 2.012\text{ V}$$

Resistor values of $R_1 = 100\ \Omega$ and $R_2 = 16.57$ kΩ will satisfy the requirements. The crossover voltages are thus $V_{TH} = 2.03$ V and $V_{TL} = 1.97$ V.

Trade-offs: If we use standard-valued resistors $R_1 = 120\,\Omega$ and $R_2 = 20\,\text{k}\Omega$, the hysteresis width is

$$\begin{aligned} V_{TH} - V_{TL} &= \left(\frac{R_1}{R_1 + R_2}\right)(V_H - V_L) \\ &= \left(\frac{0.12}{0.12 + 20}\right)[5 - (-5)] \to 59.6\,\text{mV} \end{aligned}$$

If we are able to use a reference voltage of 2.012 V, then the switching voltage is

$$V_S = \left(\frac{R_2}{R_1 + R_2}\right)V_{\text{REF}} = \left(\frac{20}{0.12 + 20}\right)(2.012) = 2.0\,\text{V}$$

Resistor tolerances will also affect the results, but will not be considered here.

Comment: In this case, the output chatter effect is eliminated for noise signals with amplitudes lower than 30 mV. The hysteresis width can be adjusted up or down to fit specific application requirements in which the noise signal is larger or smaller than that given in this example.

EXERCISE PROBLEM

Ex 15.7: Redesign the street light control circuit shown in Figure 15.32(a) such that the switching voltage is $V_S = 1$ V and the hysteresis width is 100 mV. Assume $V_H = +10$ V and $V_L = -10$ V. Also, find R such that $I = 200\,\mu$A when $v_O = V_H$. Assume $V_{BE}(\text{on}) = 0.7$ V and $V_\gamma = 0.7$ V, and assume the relay switch resistance is 100 Ω. (Ans. $R_2/R_1 = 199$, $V_{\text{REF}} = 1.005$ V, $R = 42.9$ kΩ)

Test Your Understanding

TYU 15.7 A noninverting Schmitt trigger is shown in Figure 15.29(a) Its saturated output voltages are ±12 V. Design the circuit to obtain ±200 mV crossover voltages. The maximum resistance value is to be 200 kΩ. (Ans. Set $R_2 = 200$ kΩ, then $R_1 = 3.33$ kΩ)

TYU 15.8 For the Schmitt trigger in Figure 15.30(a), the parameters are: $V_{\text{REF}} = 2$ V, $V_H = 10$ V, $V_L = -10$ V, $R_1 = 1$ kΩ, and $R_2 = 10$ kΩ: (a) Determine V_S, V_{TH}, and V_{TL}. (b) Let v_I be a triangular wave with a zero average voltage, a 10 V peak amplitude, and a 10 ms period. Sketch v_O versus time over two periods. Label the appropriate voltages and times. (Ans. (a) $V_S = 1.82$ V, $V_{TH} = 2.73$ V, $V_{TL} = 0.91$ V)

TYU 15.9 Consider the Schmitt trigger in Figure 15.31(a). Let $V_H = 9$ V and $V_L = -9$ V. Design the circuit such that $V_S = -2$ V and the hysteresis width is 0.5 V. The minimum resistance is to be 10 kΩ. (Ans. Set $R_1 = 10$ kΩ, then $R_2 = 360$ kΩ, $V_{\text{REF}} = -1.946$ V)

15.4 NONSINUSOIDAL OSCILLATORS AND TIMING CIRCUITS

Objective: • Analyze and design multivibrator circuits that provide signals with particular waveforms.

Many applications, especially digital electronic systems, use a nonsinusoidal square-wave oscillator to provide a clock signal for the system. This type of oscillator is called an astable multivibrator. In other applications, a single pulse of known height and width is used to initiate a particular set of functions. This type of oscillator is called a monostable multivibrator. First, we will examine the Schmitt trigger connected as an oscillator. Then we will analyze the 555 timer circuit. Although used extensively in digital electronic systems, these circuits are included here as comparator circuit applications.

15.4.1 Schmitt Trigger Oscillator

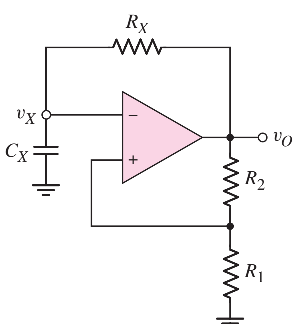

Figure 15.35 Schmitt trigger oscillator

The Schmitt trigger can be used in an oscillator circuit to generate a square-wave output signal. This is accomplished by adding an RC network to the negative feedback loop of the Schmitt trigger as shown in Figure 15.35. As we will see, this circuit has no stable states. It is therefore called an **astable multivibrator.**

Initially, we set R_1 and R_2 equal to the same value, or $R_1 = R_2 \equiv R$. We assume that the output switches symmetrically about zero volts, with the high saturated output denoted by $V_H = V_P$ and the low saturated output denoted by $V_L = -V_P$. If v_O is low, or $v_O = -V_P$, then $v_+ = -(\frac{1}{2})V_P$. When v_X drops just slightly below v_+, the output switches high so that $v_O = +V_P$ and $v_+ = +(\frac{1}{2})V_P$. The $R_X C_X$ network sees a positive step-increase in voltage, so capacitor C_X begins to charge and voltage v_X starts to increase toward a final value of V_P.

The general equation for the voltage across a capacitor in an RC network is

$$v_X = v_{\text{Final}} + (v_{\text{Initial}} - v_{\text{Final}})\, e^{-t/\tau} \tag{15.82}$$

where v_{Initial} is the initial capacitor voltage at $t = 0$, v_{Final} is the final capacitor voltage at $t = \infty$, and τ is the time constant. We can now write

$$v_X = V_P + \left(-\frac{V_P}{2} - V_P\right)e^{-t/\tau_x} \tag{15.83(a)}$$

or

$$v_X = V_P - \frac{3V_P}{2}e^{-t/\tau_x} \tag{15.83(b)}$$

where $\tau_x = R_X C_X$. Voltage v_X increases exponentially with time toward a final voltage V_P. However, when v_X becomes just slightly greater than $v_+ = +(\frac{1}{2})V_P$, the output switches to its low state of $v_O = -V_P$ and $v_+ = -(\frac{1}{2})V_P$. The $R_X C_X$ network sees a negative step change in voltage, so capacitor C_X now begins to discharge and voltage v_X starts to decrease toward a final value of $-V_P$. We can now write

$$v_X = -V_P + \left[+\frac{V_P}{2} - (-V_P)\right]e^{-(t-t_1)/\tau_x} \tag{15.84(a)}$$

or

$$v_X = -V_P + \frac{3V_P}{2} e^{-(t-t_1)/\tau_x} \tag{15.84(b)}$$

where t_1 is the time at which the output switches to its low state. The capacitor voltage then decreases exponentially with time. When v_X decreases to $v_+ = -(\frac{1}{2})V_P$, the output again switches to its high state. The process continues to repeat itself, which means that this positive-feedback circuit oscillates producing a square-wave output signal. Figure 15.36 shows the output voltage v_O and the capacitor voltage v_X versus time.

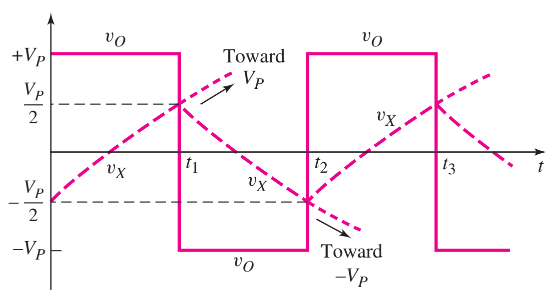

Figure 15.36 Output voltage and capacitor voltage versus time for Schmitt trigger oscillator

Time t_1 can be found from Equation (15.83(b)) by setting $t = t_1$ when $v_X = V_P/2$, or

$$\frac{V_P}{2} = V_P - \frac{3V_P}{2} e^{-t_1/\tau_x} \tag{15.85}$$

Solving for t_1, we find that

$$t_1 = \tau_x \ln 3 = 1.1 R_X C_X \tag{15.86}$$

From a similar analysis using Equation (15.84(b)), we find that the difference between t_2 and t_1 is also $1.1R_X C_X$; therefore, the period of oscillation T is

$$T = 2.2 R_X C_X \tag{15.87}$$

and the frequency of oscillation is

$$f = \frac{1}{T} = \frac{1}{2.2 R_X C_X} \tag{15.88}$$

As an example of an application of this circuit, a variable frequency oscillator is created by letting R_X be a variable resistor.

The **duty cycle** of the oscillator is defined as the percentage of time that the output voltage v_O is in its high state. For the circuit just considered, the duty cycle is 50 percent, as seen in Figure 15.36. This is a result of the symmetrical output voltages $+V_P$ and $-V_P$. If asymmetrical output voltages are used, then the duty cycle changes from the 50 percent value.

DESIGN EXAMPLE 15.8

Objective: Design a Schmitt trigger oscillator for a specified frequency.

Specifications: Assume that an ideal comparator is available. Use standard-valued resistors and capacitors in the final design.

Consider the oscillator in Figure 15.35. Design the circuit to oscillate at $f_o = 1$ kHz.

Solution: Using Equation (15.88), we can write

$$R_X C_X = \frac{1}{2.2 f_o} = \frac{1}{2.2(10^3)} = 4.55 \times 10^{-4}$$

If $C_X = 0.1\,\mu\text{F}$, then $R_X = 4.55\,\text{k}\Omega$.

Trade-offs: Using standard-valued elements with values of $C_X = 0.082\,\mu\text{F}$ and $R_X = 5.6\,\text{k}\Omega$ produces an oscillation frequency of 990 Hz, within 1% of the specified value. If element tolerance values are taken into account, a potentiometer may have to be used to produce the 1000 Hz oscillation frequency.

Comment: A larger frequency of oscillation can easily be obtained by using a smaller capacitor value.

EXERCISE PROBLEM

***Ex 15.8:** For the Schmitt trigger oscillator in Figure 15.35, the saturation output voltages are +10 V and −5 V. $R_1 = R_2 = 20\,\text{k}\Omega$, $R_X = 50\,\text{k}\Omega$, and $C_X = 0.01\,\mu\text{F}$. Determine the frequency of oscillation and the duty cycle. Sketch v_O and v_X versus time over two periods of the oscillation. (Ans. $f = 866$ Hz, duty cycle = 39.7%)

15.8 SUMMARY

- This chapter has presented several applications of op-amps and comparators that may be fabricated as integrated circuits.
- A comparator is essentially an op-amp operated in an open-loop configuration. The output signal is either a high or low saturated voltage.
- A Schmitt trigger uses a comparator with positive feedback, which produces a hysteresis in the voltage transfer characteristics. This circuit, with its hysteresis characteristic, can eliminate the chatter effect in an output signal during switching applications in which noise is superimposed on the input signal. Astable and monostable multivibrators were considered.

CHECKPOINT

After studying this chapter, the reader should have the ability to:

✓ Design a basic Schmitt trigger circuit.
✓ Design a Schmitt trigger square-wave oscillator

REVIEW QUESTIONS

3. Sketch the circuit and characteristics of a basic inverting Schmitt trigger.
4. What is meant by bistable and astable circuits?
5. What is the primary advantage of a Schmitt trigger circuit.
6. Sketch the circuit and explain the operation of a Schmitt trigger oscillator.

PROBLEMS

Section 15.3 Schmitt Trigger Circuits

15.42 For the comparator in the circuit in Figure 15.26(a), the output saturation voltages are ± 9 V. Let $V_{REF} = 5$ V. Let R_2 be a fixed resistor in series with a potentiometer. Design the circuit such that the crossover voltage can easily be varied over the range of -2 V to -4 V. The minimum resistance is to be 20 kΩ.

15.43 Consider the Schmitt trigger shown in Figure 15.29(a). Assume the saturated output voltages are $V_H = +10$ V and $V_L = -10$ V. The range of the input voltage is $-5 \le v_I \le 5$ V. Design the circuit such that the hysteresis width is 0.4 V and the maximum current in any resistor is 0.20 mA.

15.44 The saturated output voltages of the Schmitt trigger in Figure 15.28(a) are $V_H = +9$ V and $V_L = -9$ V. The range of the input voltage is $-9 \le v_I \le 9$ V. (a) Design the circuit whose minimum resistance is 2 kΩ and such that the hysteresis width is 0.2 V. (b) Using the results of part (a), determine the maximum current in the resistors.

15.45 A Schmitt trigger circuit is shown in Figure 15.28(a). The parameters are $V_H = +10$ V, $V_L = -10$ V, $R_1 = 2$ kΩ, and $R_2 = 48$ kΩ. (a) Determine the crossover voltages V_{TH} and V_{TL}. (b) Assume a sinusoidal voltage $v_I = 10 \sin[2\pi\,(60)\,t]$ V is applied at the input. During the period $33.3 \le t \le 50$ ms, determine the time periods that the output is high and the time periods that the output is low.

15.46 Consider the Schmitt trigger in Figure P15.46. Assume the saturated output voltages are $\pm V_P$. (a) Derive the expression for the crossover voltages V_{TH} and V_{TL}. (b) Let $R_A = 10$ kΩ, $R_B = 20$ kΩ, $R_1 = 5$ kΩ, $R_2 = 20$ kΩ, $V_P = 10$ V, and $V_{REF} = 2$ V. (a) Find V_{TH} and V_{TL}. (b) Sketch the voltage transfer characteristics.

15.47 The saturated output voltages are $\pm V_P$ for the Schmitt trigger in Figure P15.47. (a) Derive the expressions for the crossover voltages V_{TH} and V_{TL} (b) If $V_P = 12$ V, $V_{REF} = -10$ V, and $R_3 = 10$ kΩ, find R_1 and R_2 such that the switching point is $V_S = -5$ V and the hysteresis width is 0.2 V. (c) Sketch the voltage transfer characteristics.

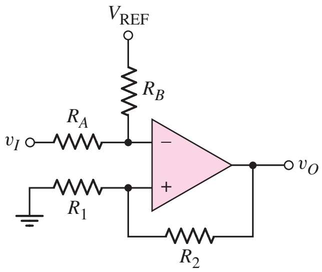

Figure P15.46

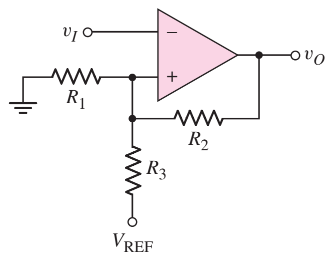

Figure P15.47

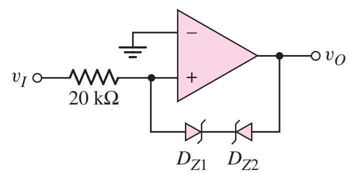

Figure P15.48

15.48 (a) Plot the voltage transfer characteristics of the comparator circuit in Figure P15.48 assuming the open-loop gain is infinite. Let the reverse Zener voltage be $V_Z = 5.6$ V and the forward diode voltage be $V_\gamma = 0.6$ V. (b) Repeat part (a) for an open-loop gain of 10^3. (c) Repeat part (a) for 2.5 V applied to the inverting terminal of the comparator.

15.49 Consider the Schmitt trigger in Figure 15.30(a). (a) Derive the expression for the switching point and crossover voltages as given in Equations (15.76) and (15.77). (b) Let $V_H = +12$ V and $V_L = -12$ V. The minimum resistance is to be 4 kΩ. Determine R_1, R_2, and V_{REF} such that the crossover voltages are $V_{TH} = -1.5$ V and $V_{TL} = -2$ V. (c) What are the currents in the resistors when (i) $v_O = V_H$ and (ii) $v_O = V_L$?

15.50 Consider the Schmitt trigger in Figure 15.31(a). (a) Derive the expressions for the switching point and crossover voltages, as given in Equations (15.78) and (15.79). (b) Let $V_H = 12$ V, $V_L = -12$ V, and $R_2 = 20$ kΩ. Determine R_1 and V_{REF} such that $V_{TH} = -1$ V and $V_{TL} = -2$ V.

15.51 Consider the Schmitt trigger in Figure P15.51. The saturated output voltages of the op-amp are $V_H = +10$ V and $V_L = -10$ V. Assume the diode turn-on voltage is 0.7 V. The range of the input voltage is $-2 \le v_I \le +2$ V. (a) Determine the crossover voltages. (b) Sketch the voltage transfer characteristics. (c) Determine I_{D1}, I_{D2}, I_{R3}, and I_{R2} for (i) $v_I = 2$ V and (ii) $v_I = -2$ V.

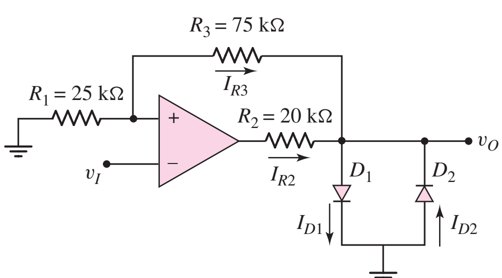

Figure P15.51

15.52 The saturated output voltages of the comparator in the circuit shown in Figure 15.33 are ±10 V. Assume forward diode voltages of 0.7 V and reverse Zener voltages of 5.6 V. The range of the input voltage is $-2 \le v_I \le +2$ V. (a) Find R_1 and R_2 such that the hysteresis width is 0.6 V. The minimum resistance value is to be 4 kΩ. (b) Find R such that the maximum current through the diodes is 0.8 mA.

15.53 Consider the Schmitt trigger with limiter, as shown in Figure 15.34. Assume the forward diode turn-on voltage V_γ is 0.7 V. (a) Determine V_{REF} such that the bistable output voltages at $v_I = 0$ are ± 5 V. (b) Find values of R_1 and R_2 such that the crossover voltages are ± 0.5 V. (c) Taking R_1, R_2, and the 100 kΩ resistors into account, find v_O when $v_I = 10$ V.

15.54 Consider the inverting Schmitt trigger with limiting network, as shown in Figure 15.34(a). Show that the crossover voltages are those given in Figure 15.34(b).

15.55 (a) For the Schmitt trigger with limiter in Figure P15.55(a), find the two output voltage values at $v_I = 0$ and the two crossover voltages. (b) Derive the expression for the slope of v_O versus v_I for $v_I > V_{TH}$.

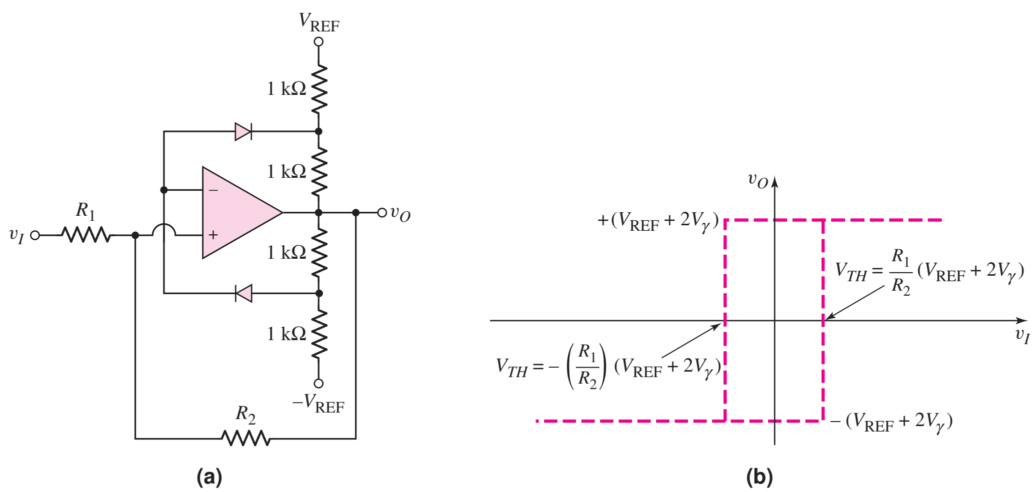

Figure P15.55

Section 15.4 Nonsinusoidal Oscillators and Timing Circuits

15.56 Consider the Schmitt trigger oscillator in Figure 15.35. The circuit parameters are $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$, $R_X = 40\,\text{k}\Omega$, and $C_X = 0.02\,\mu\text{F}$. The saturated output voltages are ± 5 V. (a) Write the expressions for v_X as a function of time assuming that v_O has switched to its high state at $t = 0$. (b) Determine the frequency of oscillation and the duty cycle.

15.57 Repeat Problem 15.56 for saturated output voltages of $V_H = +5$ V and $V_L = -10$ V.

D15.58 Design the Schmitt trigger circuit in Figure 15.35 to produce a square-wave output signal at a frequency of $f_o = 12$ kHz and a 50 percent duty cycle. Choose standard component values.

15.59 Consider the circuit in Figure P15.59. The saturated output voltages of the Schmitt trigger comparator are ± 10 V. Assume that at $t = 0$, output v_{o1} switches from its low state to its high state and C_Y is uncharged. Plot v_{o1} and v_o versus time over two periods of oscillation.

15.60 The saturated output voltages of the comparator in Figure P15.60 are ± 10 V. (a) Find R_x such that the frequency of oscillation is 500 Hz when the potentiometer is connected to point A. (b) Using the results of part (a), determine the oscillator frequency when the potentiometer is connected to point B.

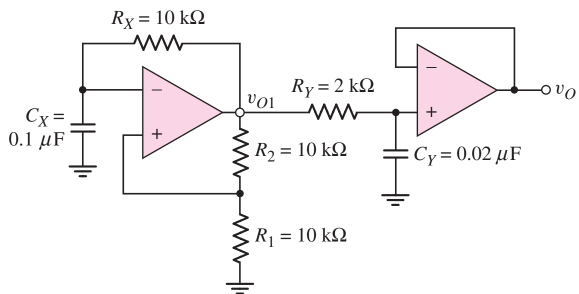

Figure P15.59

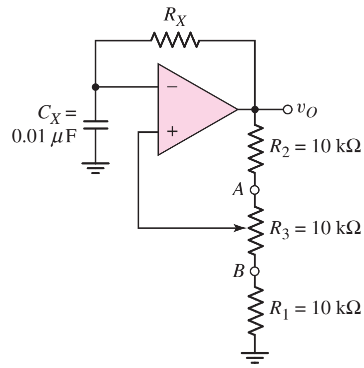

Figure P15.60